

Refutation of TLMC Conjecture 00000004225

Higher Dirichlet–Neumann zero-multiplicity identity

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The conjecture

Definition (as stated). The higher Dirichlet–Neumann operator is the boundary response operator on k -chains.

Conjecture. The multiplicity of its zero eigenvalue equals exactly the count of k -spanning trees of the boundary, and the multiplicity is one if and only if the boundary is a sphere triangulation.

Standard reading and why it fails

On a boundary complex $B = \partial M$, the combinatorial higher Dirichlet–Neumann (boundary-response) operator on k -chains is

$$\Delta_k = \partial_{k+1} \partial_{k+1}^\top + \partial_k^\top \partial_k,$$

the Hodge Laplacian. Its zero eigenspace is the space of harmonic k -chains, so

$$\text{mult}_k(B) = \dim \ker \Delta_k(B) = \beta_k(B) = f_k - \text{rank } \partial_k - \text{rank } \partial_{k+1}.$$

The simplicial spanning-tree count $\tau_k(B)$ is a *determinantal* quantity (Kalai’s matrix-tree theorem: a weighted product of the *nonzero* Laplacian eigenvalues), not a kernel dimension. Hence $\text{mult}_k = \tau_k$ fails generically, and the “multiplicity one iff sphere” clause fails in both directions.

Counterexample A: $\partial\Delta^3$ (a sphere), $k = 1$

The boundary of the tetrahedron has $(f_0, f_1, f_2) = (4, 6, 4)$, $\chi = 2$: a triangulation of S^2 . Direct computation gives

$$\Delta_1 = \partial_1^\top \partial_1 + \partial_2 \partial_2^\top = 4I_6, \quad \det \Delta_1 = 4096,$$

so $\text{mult}_1 = 0$. But $\tau_1 = \tau(K_4) = 16$ (Cayley). Thus $\text{mult}_1 \neq \tau_1$ (first clause fails), and a sphere boundary has $\text{mult}_1 \neq 1$ (the “if” direction of clause 2 fails).

Counterexample B: the 7-vertex torus T^2 , $k = 2$

The minimal triangulation of the torus (the boundary of a solid torus) has $(f_0, f_1, f_2) = (7, 21, 14)$, $\chi = 0 \neq 2$: not a sphere. Since there are no 3-faces, $\Delta_2 = \partial_2^\top \partial_2$ on $C_2 \cong \mathbb{Q}^{14}$. An explicit harmonic 2-cycle $c = (-1, \dots, -1, +1, \dots, +1)$ lies in the kernel, and a dual-tree elimination on the 21 edge equations shows the kernel is exactly $\text{span}\{c\}$; hence $\text{mult}_2 = \beta_2(T^2) = 1$. A non-sphere boundary therefore attains multiplicity 1 (the “only if” direction fails). Moreover $\tau_2 = 0$ (a 2-tree would require $\text{rank } \partial_2 = f_2 = 14$, but $\text{rank } \partial_2 = 13$), so clause 1 fails again: $1 \neq 0$.

Formalization

All objects are defined concretely in `Main.lean` (namespace `Submission00000004225`): `dnOperator`, `zeroMult`, `FinSurface` with `IsTwoSphere`, the incidence matrices `tetB1`, `tetB2`, `torusB2`, and the tree count `tetTreeCount`. The headline theorem `disproof_00000004225` states

$$\begin{aligned} & \text{zeroMult}(\text{tetDN1}) \neq \text{tetTreeCount} \\ & \wedge (\text{tetBoundary.IsTwoSphere} \wedge \text{zeroMult}(\text{tetDN1}) \neq 1) \\ & \wedge (\neg \text{torusBoundary.IsTwoSphere} \wedge \text{zeroMult}(\text{torusDN2}) = 1), \end{aligned}$$

refuting both clauses and both directions. Every proof is axiom-audited in `Check.lean`; only the ambient axioms `propext`, `Classical.choice`, `Quot.sound` appear. Numerical verification is in `reproduce.py` (sympy, exact arithmetic).